

Model	Dataset	Split Method	Metric	Best Average	Shuffled Features	MoLFormer Embeddings	 Drug & Cell Line Features	 One-Hot Encoding	 Percent Deviation (%)
DeepSynergy	O'Neil	LPO	MSE	349.58	272.33 ± 13.31	275.87 ± 13.56	252.55 ± 12.46	250.25 ± 13.24	-0.91 ± 1.76
			PCC	-	0.70	0.70	0.72	0.73	1.39
			SCC	-	0.68	0.68	0.72	0.73	1.39
DeepDDS	O'Neil	LTO	ROC AUC	0.80	0.93	-	0.93	0.94	1.08
			PR AUC	-	0.93	-	0.93	0.94	1.08
			ACC	-	0.85	-	0.85	0.86	1.18
Matchmaker	DrugComb	LPO	MSE	174.57	97.32 ± 1.13	96.76 ± 1.12	97.23 ± 1.14	100.25 ± 1.15	3.11 ± 0.54
			PCC	-	0.76	0.76	0.76	0.75	-1.32
			SCC	-	0.71	0.71	0.72	0.70	-2.78
MARSY	DrugComb	LPO	MSE	58.37	33.71 ± 1.23	33.14 ± 1.14	32.50 ± 1.20	30.42 ± 1.11	-6.4 ± 1.49
			PCC	-	0.87	0.87	0.87	0.88	1.15
			SCC	-	0.74	0.73	0.74	0.75	1.35
JointSyn	O'Neil	LTO	MSE	185.92	78.69 ± 2.32	-	76.93 ± 2.21	77.38 ± 2.21	0.59 ± 1.03
			PCC	-	0.88	-	0.89	0.88	-1.12
			R ²	-	0.77	-	0.78	0.78	0.00